

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
C01_AT2 — DATA INTERPRETATION EXERCISE

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO1 –Apply (BL3)	Assessment:	Assessment Tool 2 – Data Interpretation
Weightage:	20% of CIA	Total Marks:	20(2 Questions × 20 Marks)
CO1 Statement:	<i>Understand the fundamental concepts, principles, and techniques of data visualization.(BL2)</i>		
Student Name:	Sania Herbert	Reg. No.:	192324062
Section / Batch:	2023/AI&DS	Date:	04.08.26

Code of Conduct

I Sania Herbert(192324062) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sania Herbert

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.

Annexure A – DATA INTERPRETATION

Rubrics for Calculation **ASSESSMENT RUBRIC**
AT2 — Data Interpretation Exercise

CO1 · BT3: Apply ·

This rubric is used to assess student responses to the AT2 Data Interpretation Exercise problems, in which students interpret a described data visualization (data types, aesthetic mappings, scales, coordinate systems, and color scales) and answer accompanying sub-questions.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of data types (nominal, ordinal, numerical), aesthetic mappings, scales, coordinate systems, and color scales as they apply to the given visualization scenario.	Good understanding of the relevant visualization concepts, with only minor gaps in identifying data types, aesthetic mappings, or scale choices.	Basic understanding of visualization concepts, but with some misconceptions about data types, aesthetics, coordinate systems, or color scales.	Poor understanding of core visualization concepts; major misconceptions about data types, aesthetics, coordinate systems, or color scales.
Explanation	25%	Provides detailed, well-reasoned explanations for each sub-question, using specific details from the scenario and correct visualization terminology.	Provides clear explanations with adequate justification, referencing the scenario appropriately for most sub-questions.	Explanations are limited or superficial; lacks depth or fails to consistently connect reasoning to the given scenario.	Explanations are poor, unclear, or missing; answers are unsupported assertions with no reasoning shown.
Application	20%	Effectively applies visualization concepts to interpret the specific scenario, drawing accurate, well-reasoned conclusions from the described chart or data pattern.	Applies concepts to interpret the scenario correctly, with only minor errors in reasoning or in the conclusions drawn.	Limited application of concepts to the scenario; conclusions are weakly supported or only partially correct.	Fails to apply concepts to the scenario, or the conclusions drawn are incorrect or irrelevant to the data shown.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Organization	15%	Answers are well-structured, addressing each sub-question (a, b, c) in a clear, logical sequence with coherent flow throughout.	Answers are organized and address each sub-question, with only minor issues in flow or clarity.	Basic structure; sub-questions are addressed but answers lack clarity or a logical sequence.	Poorly organized; answers are incoherent, and sub-questions are left unaddressed or answered out of order.
Accuracy	10%	All parts of the answer — data type identification, aesthetic/scale justification, and final interpretation — are completely accurate.	Mostly accurate, with only minor omissions or a small error in one part of the answer.	Partially correct; some key points (e.g., data type, aesthetic justification, or interpretation) are missing or incorrect.	Answers are largely incorrect, with major gaps in identifying data types, aesthetics, scales, or drawing conclusions.

Scoring Guide

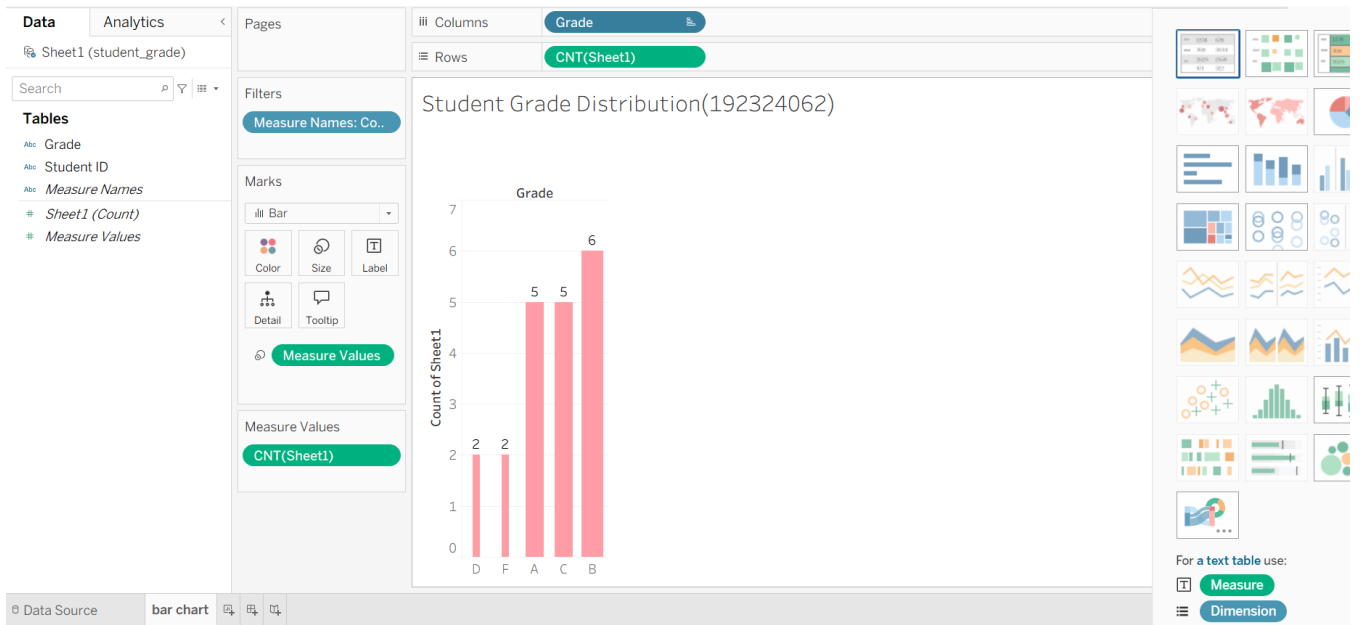
For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 5 (10 Marks)

Scenario: A university tracks student grades as letter grades (A, B, C, D, F) and plots the count of students receiving each grade as an ordered bar chart, bars arranged from F to A left to right, using a color gradient from red (F) to green (A).

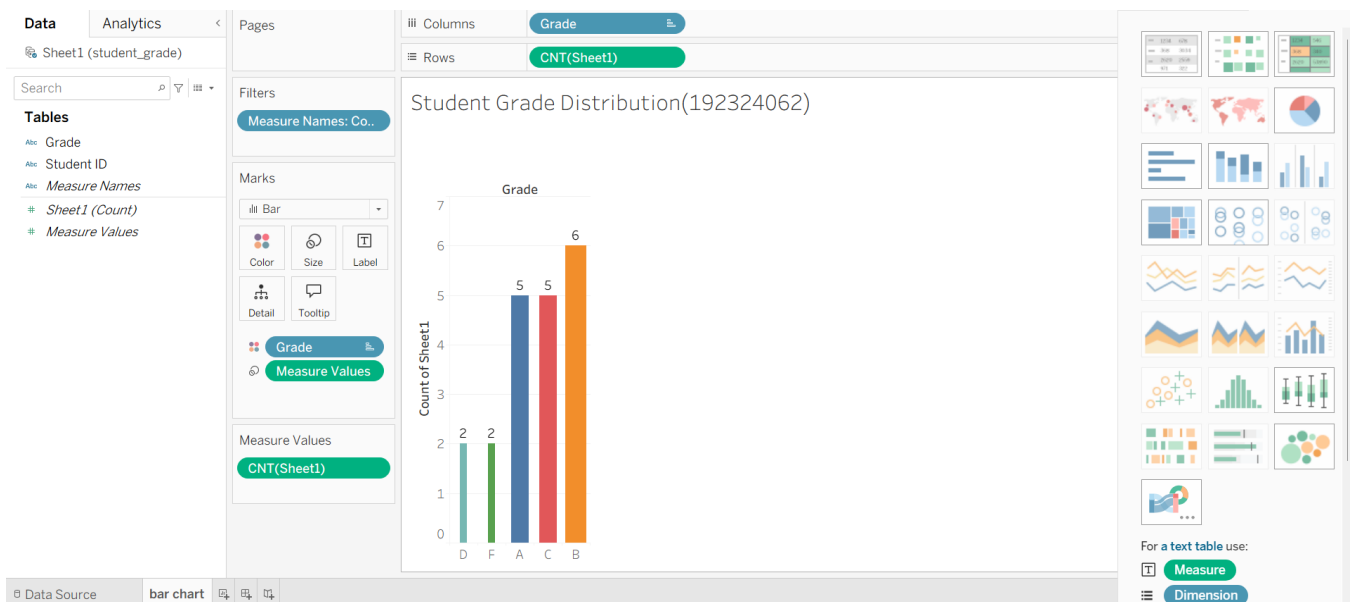
- What type of data is “letter grade,” and why does its order matter for this visualization?



Letter grades are Ordinal Categorical Data because they have a meaningful order ($F < D < C < B < A$).

b) Evaluate whether using a red-to-green gradient is an effective and accessible aesthetic choice.

The Red–Green gradient effectively represents poor-to-good performance. However, it may not be fully accessible for users with red–green color blindness. Adding labels or using a color-blind-friendly palette improves accessibility.



c) If most bars cluster around B and C, what does that indicate about the shape of the overall distribution?

Since most students received **B and C**, the distribution is concentrated around the middle grades, indicating an approximately bell-shaped (normal-like) distribution.

Topic 3: Color Scales

Problem 15 (10 Marks)

Scenario: A company profit/loss dashboard uses a red-white-blue diverging color scale, where red

represents losses, white represents breakeven, and blue represents profits.

a) Why is a diverging color scale, rather than a sequential one, appropriate for this data?

A **diverging color scale** is appropriate because the profit/loss data has a meaningful central value (**0**, which represents break-even). Values less than zero (losses) are shown in **red**, values equal to zero (break-even) are shown in **white**, and values greater than zero (profits) are shown in **blue**. This makes it easy to distinguish negative and positive financial performance. A sequential color scale is not suitable because it only shows values progressing from low to high without emphasizing the midpoint.

b) If a month's cell appears as a very dark blue, what does that indicate about that month's financial performance?

A very dark blue cell indicates that the company earned a high profit during that month. In the diverging color scale, darker shades of blue represent larger positive values, meaning the month's financial performance was significantly above the break-even point.

c) What data value would be represented by white, and why is that meaningful for this specific variable?

White represents a **profit value of 0**, which means the company **broke even**—it neither made a profit nor incurred a loss. This is meaningful because it serves as the midpoint of the diverging color scale, clearly separating months with losses (red) from months with profits (blue).

